

Battery Placement and Weight Distribution

Problem

A 7.4 V LiPo battery has a manageable weight, but its physical dimensions create a significant packaging challenge. Its size may negatively affect the robot's weight distribution.

Placing the battery at the rear of the robot would shift additional mass onto the solid rear axle. This is undesirable for the Ackermann steering system, as excessive rear weight can reduce the load on the front wheels and contribute to **understeer**.

Therefore, the battery must be positioned in a way that maintains a more balanced weight distribution without significantly increasing the robot's dimensions.

Primary Solution

A possible solution is to use a smaller **3.7 V LiPo battery**, potentially around 500 mAh, combined with voltage converters:

- **3.7 V → 7.4 V converter:** Supplies the required higher voltage to the Nano, motors, and potentially other sensors.
- **3.7 V → 5 V converter:** Provides a regulated 5 V supply for the Raspberry Pi.
- **3.7 V LiPo:** Reduces the physical packaging constraints created by a larger 7.4 V battery.

This approach would allow the battery to be packaged more efficiently while maintaining the required voltage levels for the robot's electronics.

Alternative Solution

Another option is to increase the height of the robot slightly and place the battery **above the steering servo**.

Placing the battery further forward may not be practical because it would require the servo linkage to be extended significantly, potentially introducing additional flex, play, or mechanical inaccuracies.

Positioning the battery above the servo would therefore provide a compromise between:

- Maintaining a suitable front-to-rear weight distribution.
- Avoiding excessive extension of the steering linkage.

- Keeping the chassis footprint compact.
- Minimizing changes to the existing steering geometry.

The final solution will be selected based on testing of weight distribution, steering behaviour, battery performance, and overall packaging constraints.